

Fady Fawzy Youssef

Giza, Egypt

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[LinkedIn](#) — [GitHub](#)

Summary

Computer Engineering student at Cairo University with strong interests in computer networks, embedded systems, web development, and artificial intelligence. Ranked third in the preparatory engineering year with a GPA of 3.94. Passionate about hands-on engineering projects, low-level programming, and solving real-world technical problems through practical experience and teamwork.

Education

Cairo University, Faculty of Engineering	Giza, Egypt
Computer Engineering	2025 – 2029

Experience & Achievements

Power BI Trainee — Digital Egypt Pioneers Initiative	Egypt
2025 – Present	

- Learning data analysis and visualization using Power BI.
- Working with dashboards, reports, and data-driven insights.

HR Member — IEEE CUSB	Giza, Egypt
2025 – Present	

- Assisted in organizing team activities and supporting member coordination.
- Participated in HR operations, communication, and event preparation.

Research Participant, TCCD Research Day 15th Edition	2026
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- Achieved 4th place in the Computer Department.
- Presented research on *Adaptive Low-Rank Memory Compression Framework for Operating Systems*.
- Focused on improving memory efficiency through adaptive low-rank compression techniques.

Production Line Operator — EVA Pharma	Giza, Egypt
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- Operated industrial machines independently during summer training.
- Gained practical experience with production systems and teamwork in industrial environments.
- Improved communication and problem-solving skills through collaboration with technicians and supervisors.

Electrical & Technical Intern — EVA Pharma	Giza, Egypt
Jun 2025 – Oct 2025	

- Assisted engineers and technicians in electrical systems and industrial operations.
- Worked with backup generators, server/data panels, and electrical management tasks.
- Supported AutoCAD work, technical documentation, and professional email communication.
- Applied engineering knowledge in a real industrial environment and handled increasing responsibilities.

Projects

Smart Nurse Robot	2026
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[Repo](#)

- Developed a fully autonomous medical assistant robot using bare-metal ARM Thumb-2 Assembly on STM32 without HAL libraries.
- Integrated multiple peripherals and protocols including I2C, SPI, UART, ADC, PWM, Bluetooth, RFID, and ultrasonic navigation.
- Implemented real-time monitoring features such as Heart Rate, SpO2, IV drop-rate tracking, and pressure sensing.
- Built a custom TFT dashboard with telemetry visualization and wireless monitoring support.

Flowchart Designer and Simulator	2025
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[Repo](#)

- Developed a complete flowchart designer and simulator in C++ without UI frameworks.
- Implemented copy/paste, undo/redo, validation, and automated C++ code generation features.

Restaurant Management System	2026
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[Repo](#)

- Developed a restaurant order management simulation system for dine-in, takeaway, and delivery orders.
- Implemented dynamic assignment algorithms for chefs, tables, and delivery scooters.

FPGA ALU Logic Design — Academic Project	2025
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- Designed and implemented an Arithmetic Logic Unit (ALU) using Verilog.
- Supported arithmetic operations, BCD-to-excess-3 conversion, and FPGA-based sequential logic.

- Simulated and verified the design on a DE1-SoC FPGA board using ModelSim.

2026

Innovative Lampshade Project

Drive

- Led a team in designing and building a recyclable PVC-based decorative lampshade.
- Applied sustainable design principles using reused materials.
- Received 3rd place and was selected as a showcase model by the department head.

SDG 9 — Industry, Innovation, and Infrastructure

- Led a team to research and present Sustainable Development Goal 9.
- Prepared presentation materials and coordinated public speaking tasks.
- Received excellent feedback for presentation quality and teamwork.

Skills & Languages

- **Programming Languages:** C, C++, Python, JavaScript, HTML, CSS, SQL, Verilog, ARM Assembly
- **Tools & Platforms:** VS Code, Git, GitHub, Keil uVision
- **Soft Skills:** Leadership, Adaptability, Teamwork, Communication, Problem Solving
- **Languages:** Arabic (Native), English (Very Good)